

Trichotillomania: A perspective from bibliometric analysis

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Abstract

Aim: Trichotillomania, an obsessive-compulsive-related disorder, is defined by the recurrent act of pulling out one's own hair from different areas of the body. Despite the considerable body of research dedicated to the subject of trichotillomania, the overarching trends that unify these studies remain obscure. The purpose of the present bibliometric analysis was to ascertain these trends.

Method: To achieve this objective, we conducted a thorough search of publications in the Web of Science database and subsequently evaluated the acquired data using VOSviewer software.

Results: The most cited article on trichotillomania was written by Simonoff et al. The most prolific writer on trichotillomania is Grant JE. The most publications on the subject of trichotillomania were published in the "Journal of Obsessive Compulsive and Related Disorders" and the most frequently repeated keyword is trichotillomania. Trichotillomania studies have focused on treatment, clinical features, and other accompanying psychiatric conditions.

Conclusion: Potential areas of research could include treatment methods in addition to the psychiatric and physical comorbidities of trichotillomania, and efforts to enhance international collaborations in this domain should be intensified.

Keywords: author, bibliometric analysis, cited, journal, trichotillomania

The psychiatric condition known as trichotillomania (TTM), which the Diagnostic and Statistical Manual of Mental Disorders, 5th edition categorizes as an obsessive-compulsive-related disorder, is characterized by the person's repetitive act of pulling out hair from any anatomical location, resulting in hair loss.¹ Although the scalp, eyebrows, eyelashes, and pubic region are the most common areas affected by hair pulling,² the extremities, armpits, and chest can also be affected on occasion.³ Despite the fact that epidemiological data pertaining to the pediatric age group is limited, a recent systematic review reveals that trichotillomania has a lifetime prevalence of 1.14% and any hair-pulling behavior is 8.84%.⁴ Various genetic,^{5,6} neuroimaging,^{7,8} and psychosocial^{9,10} investigations have been conducted in an effort to determine the etiology of TTM. TTM not only co-occurs with psychiatric disorders, including obsessive-compulsive disorder and tic disorder, but also leads to adverse consequences such as bullying from peers and isolation from society as a result of the cosmetic issues it produces.^{2,11–14}

Bibliometric analysis is one approach utilized to obtain information about prior publications; it provides

quantitative analysis by scrutinizing scientific document attributes, including but not limited to subject matter, publication year, institution of contribution, number of authors, sources consulted, and citation count.¹⁵ Bibliometric studies serve to unveil the progression of scholarly works, contribute to the identification of overarching trends in methodology and contemporary advancements in the pertinent domain, and furnish a strategic guide for subsequent researchers.¹⁶ Bibliographic analysis maps are employed to direct new research in the field of psychiatry, just as they are in numerous other scientific disciplines. A manifestation of this circumstance is the growing quantity of bibliometric investigations devoted to psychiatric disorders, with psychotic disorders, mood disorders, addiction, and anxiety disorders carrying particular weight.^{17–21}

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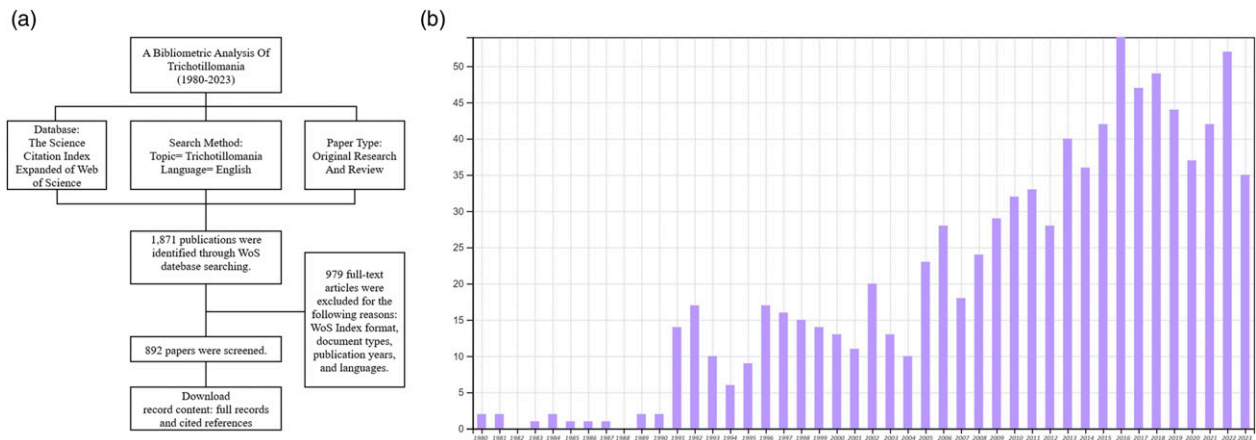


Figure 1. Research methodology for bibliometrics and distribution of articles by years. (a). The research methodology for bibliometrics. (b). Distribution of 892 articles by years.

Table 1. The top 10 articles in the field of trichotillomania

The most cited article (citations) and (authors)	Journal	DOI
Psychiatric disorders in children with autism spectrum disorders: Prevalence, comorbidity, and associated factors in a population-derived sample (2126) ²⁵	Journal Of The American Academy Of Child And Adolescent Psychiatry	10.1097/chl.0b013e318179964f
The neuropsychology of obsessive compulsive disorder: The importance of failures in cognitive and behavioral inhibition as candidate endophenotypic markers (597) ³⁸	Neuroscience And Biobehavioral Reviews	10.1016/j.neubiorev.2004.11.006
Probing compulsive and impulsive behaviors, from animal models to endophenotypes: A narrative review (454) ³⁹	Neuropsychopharmacology	10.1038/npp.2009.185
The coming of age of self-mutilation (448) ⁴⁰	Journal Of Nervous And Mental Disease	10.1097/00005.053-199,805,000-00001
Motor inhibition and cognitive flexibility in obsessive-compulsive disorder and trichotillomania (434) ⁴¹	American Journal Of Psychiatry	10.1176/appi.ajp.163.7.1282
A double-blind comparison of clomipramine and desipramine in the treatment of trichotillomania (hair pulling) (405) ⁴²	New England Journal Of Medicine	10.1056/NEJM198908243210803
Characteristics of 60 adult chronic hair pullers (366) ⁴³	American Journal Of Psychiatry	10.1176/ajp.148.3.365
The relationship of obsessive-compulsive disorder to possible spectrum disorders: Results from a family study (351) ⁴⁴	Biological Psychiatry	10.1016/S0006-3223(00)00831-3
The promise of N-acetylcysteine in neuropsychiatry (312) ⁴⁵	Trends In Pharmacological Sciences	10.1016/j.tips.2013.01.001
The neurobiology and genetics of impulse control disorders: Relationships to drug addictions (308) ⁴⁶	Biochemical Pharmacology	10.1016/j.bcp.2007.06.043

Similar to other psychiatric disorders, bibliometric analyses have been employed to ascertain trends in research on a range of disorders associated with obsessive-compulsive disorder.^{22,23} Nevertheless, despite the abundance of research devoted to TTM, one of the

obsessive-compulsive-related disorders, no investigation has been undertaken to ascertain the overarching patterns identified in these studies. Hence, we undertook this investigation to analyze the research trends on TTM by a bibliometric analysis.

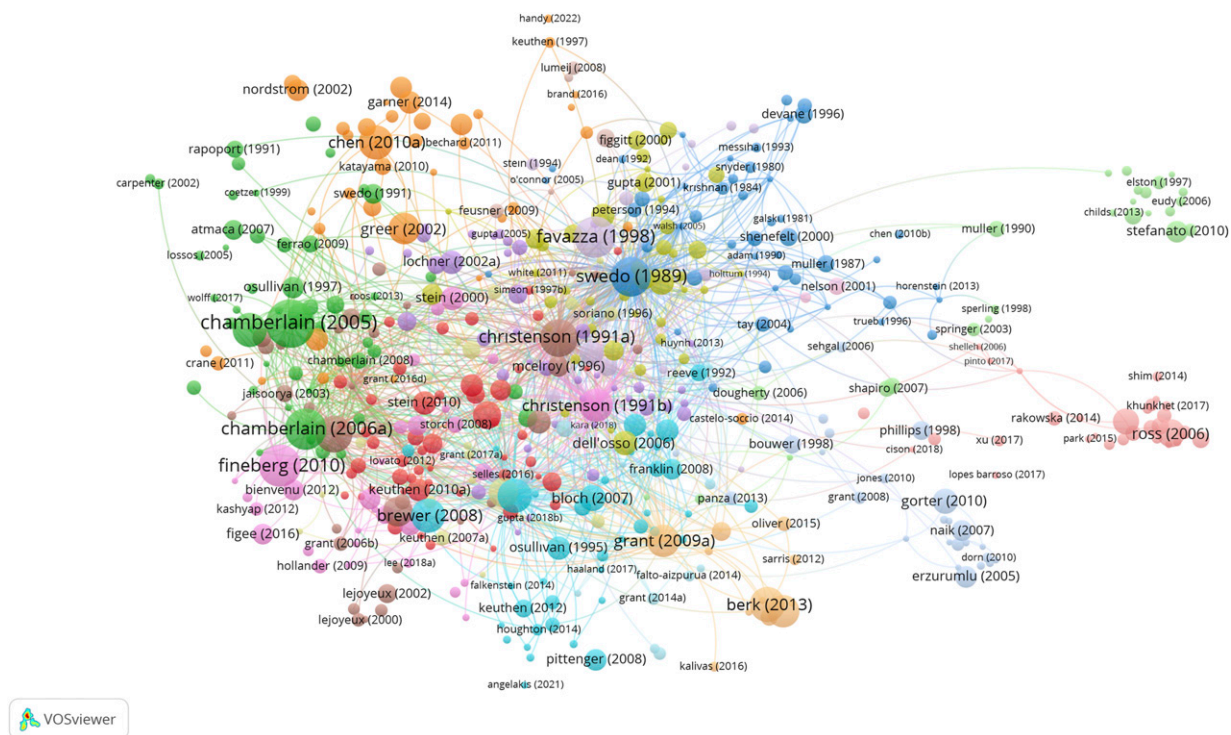


Figure 2. Citation network of publications.

Table 2. The top 10 journals in the field of trichotillomania

Journal	articles count	Country	Impact factor (5 years)
Journal Of Obsessive Compulsive And Related Disorders	57	Netherlands	1.9
Comprehensive Psychiatry	40	USA	5.1
Psychiatry Research	40	Ireland	6.2
Journal Of Clinical Psychiatry	27	USA	5.3
Journal Of Psychiatric Research	17	England	5.1
Annals Of Clinical Psychiatry	16	USA	2
Clinics In Dermatology	16	USA	3.5
Dermatologic Therapy	14	England	3.4
Depression And Anxiety	13	USA	7.5
International Journal Of Dermatology	13	USA	2.9

USA: United States of America

Methods

Figure 1(a) illustrates the research methodology for bibliometrics. The publications' data were sourced from the Science Citation Index Expanded (SCI-E) database on the Web of Science (WoS). Articles whose topic was trichotillomania and whose language of publication was English were searched in the WoS database. The data search was conducted on January 6, 2024, and publications from the establishment of the database until December 31, 2023 were included. Only research publications in the form of original articles and review articles were included,

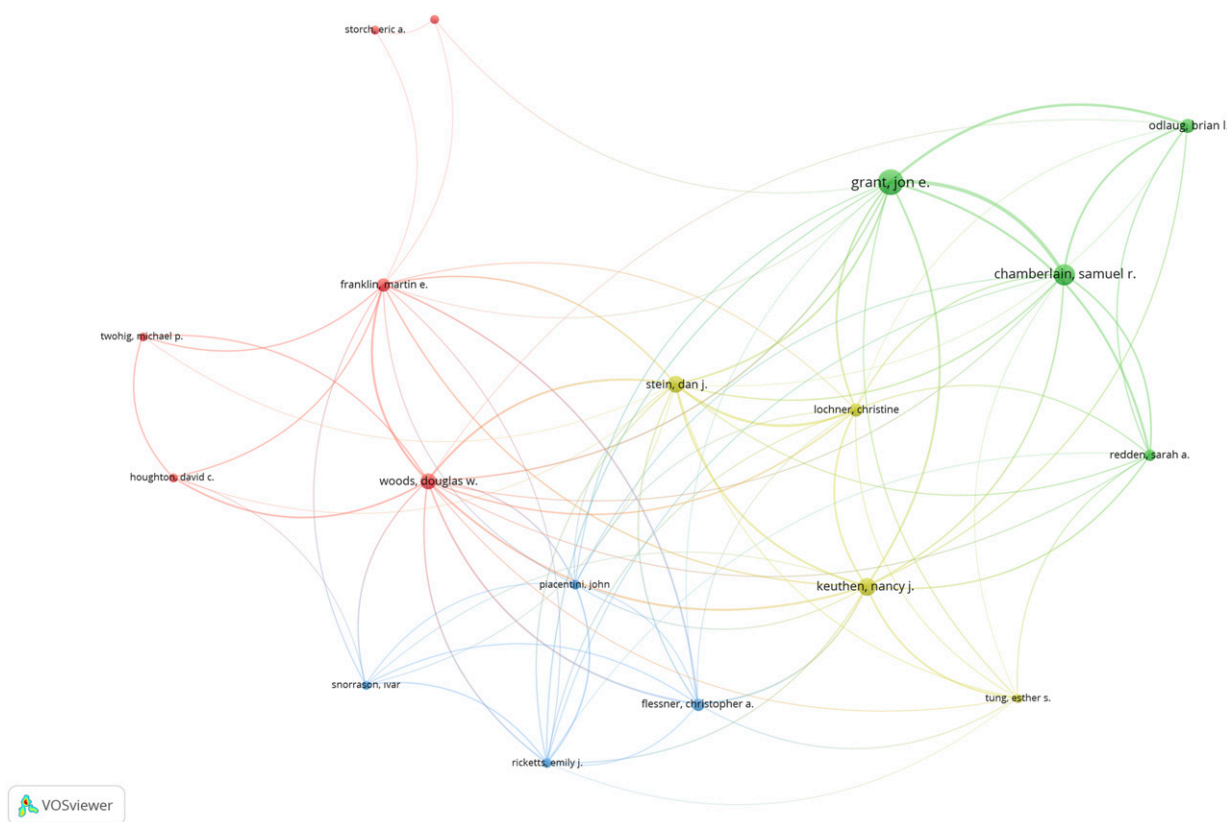
whereas other types of publications were excluded. From the publications that were included, the data were extracted, exported as a "Tab Delimited File" and subsequently recorded as "Full Records and Cited References." VOSviewer software version 1.6.20 was used for bibliometric analysis and visualize information.

Results

Upon analyzing the distribution of 892 publications (703 original research, 189 reviews) across different categories, it is evident that the majority of papers were focused on

Table 3. The top 10 authors in the field of trichotillomania

Authors	Documents	The most cited article (citations) and authors
Grant JE	93 (10.426%)	N-acetylcysteine, a glutamate modulator, in the treatment of trichotillomania: A double-blind, placebo-controlled study (276) ³⁴
Stein DJ	71 (7.960%)	The trichotillomania impact Project (TIP): Exploring phenomenology, functional impairment, and treatment utilization (222) ⁴⁷
Chamberlain SR	67 (7.511%)	The neuropsychology of obsessive compulsive disorder: The importance of failures in cognitive and behavioral inhibition as candidate endophenotypic markers (597) ³⁸
Keuthen NJ	58 (6.502%)	The Massachusetts-general-Hospital (mgh) hairpulling scale .1. Development and factor-analyses (305) ⁴⁸
Woods DW	39 (4.372%)	The trichotillomania impact project (TIP): Exploring phenomenology, functional impairment, and treatment utilization (222) ⁴⁷
Lochner C	32 (3.587%)	Childhood trauma in obsessive-compulsive disorder, trichotillomania, and controls (143) ⁴⁹
Odlaug BL	31 (3.475%)	N-acetylcysteine, a glutamate modulator, in the treatment of trichotillomania: A double-blind, placebo-controlled study (276) ³⁴
Franklin ME	26 (2.915%)	The trichotillomania impact Project (TIP): Exploring phenomenology, functional impairment, and treatment utilization (222) ⁴⁷
Flessner CA	21 (2.354%)	The trichotillomania impact Project (TIP): Exploring phenomenology, functional impairment, and treatment utilization (222) ⁴⁷
Redden SA	19 (2.130%)	N-acetylcysteine in the treatment of excoriation disorder: A randomized clinical trial (93) ⁵⁰

**Figure 3. Co-authorship network of authors.**

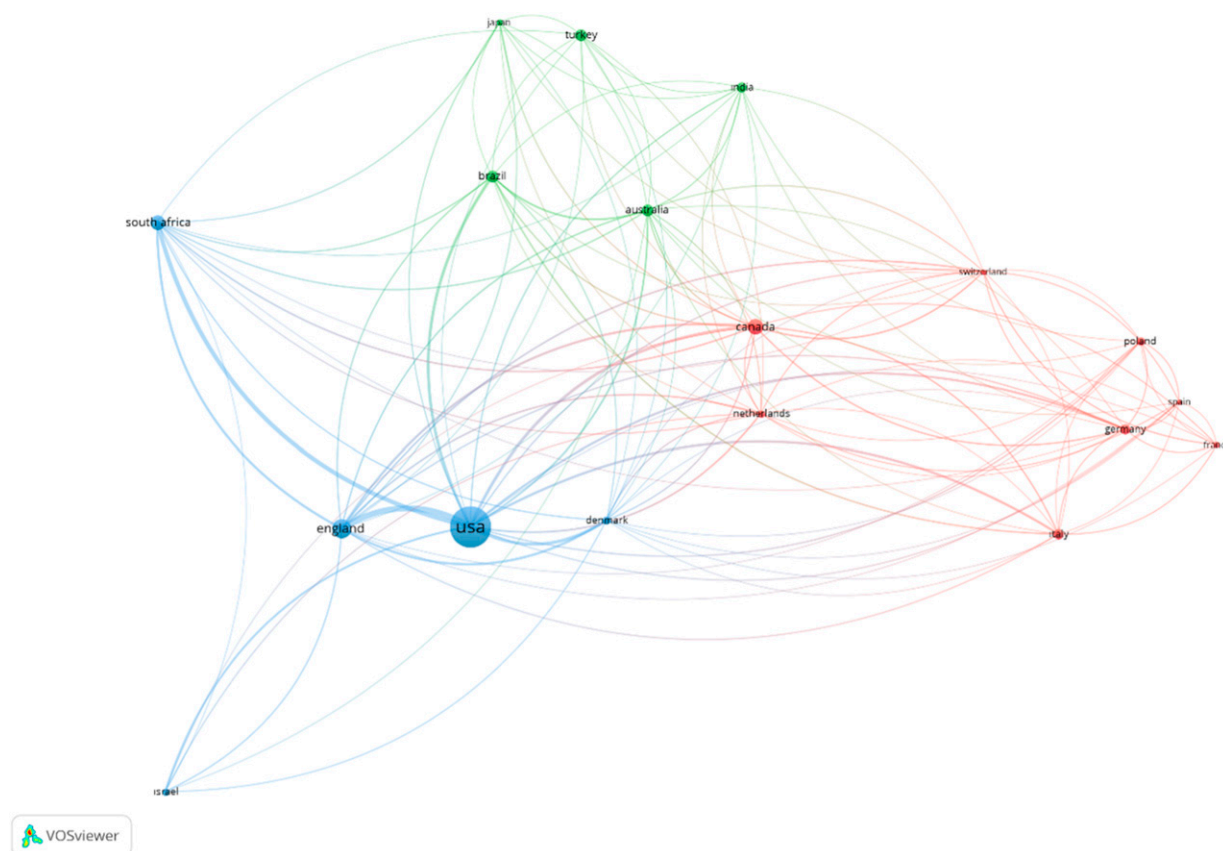


Figure 4. Co-authorship network of countries.

psychiatry (483 articles), accounting for 54.148% of the total. Following closely is the field of dermatology with 164 articles, representing 18.386%. An analysis of the publication distribution by year reveals that 2016 was the most fruitful year, producing 54 articles (6.054%), followed by 2022 with 52 articles (5.830%). [Figure 1\(b\)](#) presents the distribution of articles written on TTM by year.

Upon conducting a citation analysis of the publications, it becomes evident that the article authored by Simonoff et al. is the most frequently cited, with a total of 2126 citations. [Table 1](#) provides information on the 10 most cited articles in the field of trichotillomania. Based on the examination of 531 publications that fulfilled the requirement of having a minimum of 10 citations, the article authored by Swedo et al. has the highest number of links to other articles, with a total of 137 links ([Figure 2](#)). Swedo (1989) has 137 links, Christenson (1991a) has 122 links, Christenson (1991b) has 101 links, Keuthen (1995) has 97 links, and Woods (2006) has 24 links, respectively.

When analyzing the distribution of 892 publications throughout several journals, it becomes evident

that the “Journal of Obsessive-Compulsive and Related Disorders” had the highest number of publications, with a total of 57. [Table 2](#) provides a list of the 10 journals that have published the greatest number of articles pertaining to trichotillomania.

Data was analyzed from 2498 authors across 892 publications. Grant JE, the author with the most works on TTM, has 93 articles (10.426% of 892 publications), and his most cited work on TTM has a total of 276 citations. [Table 3](#) provides information pertaining to the 10 authors who have generated the most output in the domain of trichotillomania.

Based on the findings of the co-authorship analysis, a network map was constructed by ascertaining the most interconnected and collaborative authors through the determination of a minimum of 10 publications and 10 citation criteria. [Figure 3](#) displays inter-author cooperation through the use of co-author links. It is seen that authors with the greatest number of works are also the most connected. Woods DW has 15 links, Stein DJ has 15 links, Grant JE has 13 links, Keuthen NJ has 13 links, and Chamberlain SR has 10 links, respectively.

Co-authorship relationships between countries with a minimum of 10 publications and 10 citations are illustrated in Figure 4. The countries with the most publications and citations on TTM are the USA and the UK. It is seen that countries with the greatest number of works are also the most connected. USA has 17 links, Denmark has 17 links, Netherlands has 17 links, England has 16 links, and Australia has 16 links, respectively.

The most prevalent keyword identified through an analysis of the publications' keyword usage is "trichotillomania," which appears 321 times. After this, "obsessive-compulsive disorder" is repeated 69 times, "hair pulling" is repeated 48 times, "skin picking" is repeated 46 times, and "trichobezoar" is repeated 41 times. The network map, which was generated by employing 41 keywords that were repeated a minimum of 10 times, is illustrated in Figure 5.

Discussion

Our study is the first bibliometric study to determine the general trends observed in studies on TTM, one of the obsessive-compulsive-related disorders. In our study, we analyzed the distribution of articles written on the subject of TTM by years, the citation links between articles, co-authorship links between authors and countries, and the links between keywords.

The proliferation and variety of scientific papers across numerous scientific disciplines during the past quarter century present clear signs of our tremendous progress in the realm of science. The increasing number of studies on TTM over the past decade, as evidenced by the highest number of publications on the subject in 2016 and 2022, suggests that the aforementioned statement holds true with regard to TTM as well. TTM is classified as a psychiatric disorder, but it is also recognized as a dermatological disease because of its impact on physical appearance and potential to create other skin issues.²⁴ Given this viewpoint, it was anticipated that research on TTM would be focused in the domains of psychiatry and dermatology.

Upon analyzing the citation rates of works on TTM, it becomes evident that the publications that receive the highest number of citations are those that investigate the association between TTM and comorbidity. The study conducted by Simonoff et al., which investigated the co-occurrence of psychiatric disorders in individuals with autism, holds the highest number of citations among studies focusing on TTM.²⁵ Research has indicated that individuals with autism may exhibit behavioral issues such as hair pulling, which can lead to the development of trichobezoar.^{26,27} Furthermore, it has been demonstrated that individuals diagnosed with TTM tend to display more

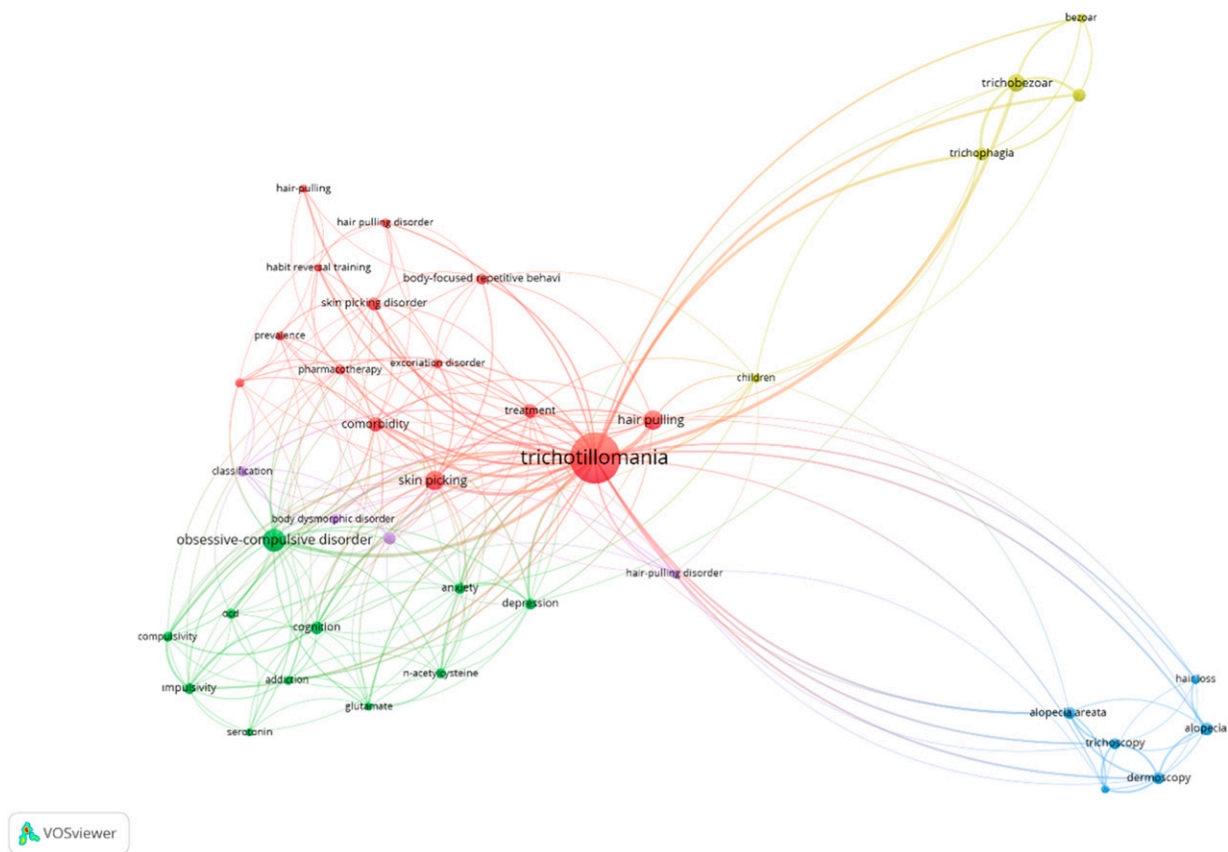


Figure 5. Co-occurrence network of keywords.

characteristics associated with autism.²⁸ It is conceivable that the popularity of articles in the TTM literature exploring the connection between TTM and autism will serve as a guide for new researchers' study methodologies.

When examining the journal distribution of publications on TTM, it is evident that the journals are distributed throughout the domains of psychiatry and dermatology, as anticipated. Furthermore, the prominence of "Depression And Anxiety" as a journal with a high number of publications on TTM underscores the connection between TTM and comorbidity in the patterns of published content. It is common knowledge that TTM is frequently observed in conjunction with emotional issues, and the severity of these two conditions exacerbates one another.^{29–31}

Before beginning to design a study on TTM, the quality of the proposed research will be enhanced in numerous ways by incorporating the insights of authors who have prior experience in the field. It is crucial to analyze the output of authors in this context. On examining the productivity of the authors of publications with a TTM theme, it becomes evident that Grant JE exhibits the highest level of productivity. An analysis of Grant JE's research on TTM reveals that publications pertaining to TTM treatment, TTM clinical characteristics, and concurrent psychiatric conditions are prominent.^{32–34} This also applies to other authors working on TTM. However, examining the authors' most recent productivity areas reveals a comparatively high number of articles about TTM treatment.

The utilization of keywords by academic databases to retrieve specific articles underscores the criticality of selecting suitable keywords during the composition of a research manuscript.³⁵ In this regard, keywords increase the article's visibility and provide the reader with information regarding its content. As anticipated, the most frequently occurring keywords associated with TTM are those that highlight the disorder's correlation with the obsessive-compulsive spectrum. Furthermore, trichobezoar is a frequently occurring keyword, which underlines the volume of literature devoted to the disorder's physical complications. Occupation of the gastrointestinal tract by trichobezoars is a potentially fatal complication of TTM; the scientific literature refers to this condition as Rapunzel Syndrome.³⁶

It is widely recognized that the initial instance of TTM was documented by the French dermatologist Hallopeau in 1889.³⁷ Including original research and reviews in English after the establishment of the Web of Science database can be seen as a limitation in our research within this specific context. Furthermore, the study's scope was limited to data exclusively sourced from the Web of Science database, without incorporating supplementary information from other databases like PubMed and Scopus. These limitations should be addressed in more extensive and sophisticated bibliometric studies on TTM.

Since TTM was first described many years ago, our comprehension of it has grown as a result of the assiduously

exerted efforts of numerous researchers. The number of research articles devoted to TTM is expanding at a rapid rate, and trends in this area of study are also evolving. Potential areas of research could include treatment methods in addition to the psychiatric and physical comorbidities of trichotillomania, and efforts to enhance international collaborations in this domain should be intensified.

Disclosure

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